KEERTI YADAV

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RESEARCH PROFILE

Recent Ph.D. graduate in Regional Development with experience collaborating across interdisciplinary research teams and with academic and professional partners. Experience spanning in spatial data analysis, transit and mobility data, and quasi-experimental causal inference for urban infrastructure and transport policy. Doctoral research applies staggered Difference-in-Differences (Callaway-Sant'Anna) methodology to estimate the heterogeneous regional effects of high-speed rail across 31 South Korean cities (2008-2020), building and validating a multi-source panel dataset that integrates administrative records, road-network infrastructure, and satellite-derived (Landsat/Sentinel-2 NDBI) land-use change parameters. My current work involves developing and applying geospatial and deep learning approaches to climate-risk assessment, environmental and geophysical data interpretation, and urban and transportation-related research, while collaborating with researchers across interdisciplinary fields. Overall research experience spans urban infrastructure, transport, regional development, and Geospatial AI, with a focus on applying spatial data and quantitative methods to real-world urban and regional challenges.

EDUCATION

Ph.D., Department of Geography **2022 - 2026**
Kongju National University, South Korea

Thesis: Heterogeneous Regional Effects of High-Speed Rail on Small and Medium-Sized Cities in South Korea: Focusing on Staggered Difference-in-Differences Analysis (2008-2020)

Language Training Program **2021 - 2022**
Busan University of Foreign Studies, South Korea

M.A., Centre for Korean Studies **2019 - 2021**
School of Language, Literature and Cultural Studies, Jawaharlal Nehru University, India

B.A., Centre for Korean Studies **2016 - 2019**
School of Language, Literature and Cultural Studies, Jawaharlal Nehru University, India

RESEARCH EXPERIENCE

Ph.D. Researcher - Regional Development & Urban Transport **2022 - 2026**
Kongju National University, South Korea

- Identified heterogeneous regional effects of high-speed rail across 31 cities (2008–2020) using a quasi-experimental research design.
- Constructed and validated an original multi-source panel dataset by systematically compiling and harmonizing socioeconomic, infrastructure, and administrative data using R and Python.
- Developed an original city-level geospatial dataset by integrating Landsat/Sentinel-2 imagery with socioeconomic and land-use data using Google Earth Engine, ArcGIS, and QGIS to assess changes in built-up areas and land use.
- Worked with public and geospatial datasets, using Python, R, OSMnx, and Kakao Maps and ACS API to collect, integrate, and analyse facilities located, road-network and transport data.
- Developed India-focused transit and mobility recommendations by comparing South Korean policies, spatial data, and governance practices with Indian contexts in collaboration with local governance bodies and academic partners.
- Kept the project well-documented and repeatable, helping collaborators follow consistent methods and achieve reliable results over time.
- Extended doctoral research into applied climate risk assessment by conducting a study on land-use vulnerability in Zabö agriculture (Northeast India) and a collaborative GIS-based deep ensemble model of drought and heatwave risk in Rajasthan.

Research Assistant - Regional Development Research Lab **2023 - 2024 (6mos)**
Kongju National University, South Korea

PUBLICATIONS

Published

- Keerti, Y., & Jin, J. H. (2025). Impact of High-Speed Rail on Urban Regions in South Korea: A Systematic Literature Review. The Geographical Journal of Korea, 59(3), 151-170. https://doi.org/10.22905/kaopqj.2025.59.3.1
- Yadav, K., Choudhary, V. A systematic approach for restoring sustainability in the hilly regions of northeast India: the Zabö conceptual framework.GeoJournal 91, 169 (2026). https://doi.org/10.1007/s10708-026-11720-6

Under first Stage Review

- Baluch, K., Nasir, M. F., Yadav, K., Qaisrani, S. A., & Wang, J. Training-Free Vision-Language Models for Near-Surface GPR Radargram Interpretation: A Benchmark of Subsurface Anomaly Detection and Reliability. Research Article.

Ongoing Research Collaborations

- Choudhary, V., Yadav, K., Mondal, M., & Punia, M. Analysing Drought and Heatwave Risk over the State of Rajasthan: Integrating Hybrid Deep Ensemble and GIS-Based Soft Computing.
- Yadav, K., & Bhattacharjee, A. Bridging the Gap: Governance Lessons from South Korea for Urban Mobility in Indian Megacities.

Independent Research

- Developing a graph neural network framework to assess how transit accessibility, walkability, and geographic connectivity shape functional urban regions across one city from 3 nations (US, Korea and India).

CONFERENCE PRESENTATIONS

- "Climate Change Vulnerability of Traditional Land Use Practices in Zabō Agriculture, India" - KAGIS Fall Conference & IAGI International Symposium, 2025
- "Exploring Rural Sustainability through AHS (Agricultural Heritage System)" - National Conference on Sustainable Rural Development, October 2023
- "Human Resource: Success Key of South Korea" - 7th AKS Young Scholars' International Webinar on Korean Studies, JNU, New Delhi, June 2020

ADDITIONAL PROFESSIONAL EXPERIENCE

Korean-English Translator & Cultural Liaison / International Program Coordinator **2020 - Present**
International forums & institutions, South Korea & India

- Supported communication across international programs as a translator and cultural liaison between Korean institutions and international/Indian delegations.
- Food Week Korea exhibition, B2B (Seoul, 2026)
- 47th International Exhibition on Environmental Technology & Green Energy, B2B (Seoul, 2026)
- International Forum on One Korea (Seoul, 2024)
- Andong Cultural Exchange (Andong, 2023)
- International exchange program for HUFS students in Delhi (2020)

Content Quality Reviewer, Media Marketing **2020 - 2021**
Match Group, Seoul, South Korea

- Planned and coordinated user-focused events by analysing user interests and engagement patterns to support relevant community experiences.
- Reviewed and quality-checked user-submitted content for compliance with platform policies and content standards, ensuring consistency and accuracy under established review guidelines.

TECHNICAL & RESEARCH SKILLS

GIS & Geospatial Analysis
ArcGIS · QGIS · Spatial data analysis · Spatial data quality assurance · Boundary and attribute validation · Multi-source geospatial data integration

Remote Sensing & Earth Observation
Google Earth Engine · Landsat · Sentinel-2 · NDBI · Land-use/land-cover change analysis · Satellite-derived variables · Earth observation data processing
Programming & Geospatial Data Engineering
Python · R · OSMnx · Automated data ingestion · Data quality control · Geospatial dataset preparation
Transport & Mobility Analytics
OpenStreetMap/OSMnx/overture maps · ACS API · HSR and transit datasets · Street-network analysis · Transit accessibility · Transport network integration · Mobility analysis
Causal Inference & Quantitative Methods
Panel data analysis · Staggered Difference-in-Differences · Callaway-Sant'Anna estimator · Quasi-experimental design · Causal inference
Urban & Regional Research
Urban infrastructure · Transport policy · Regional development · Spatial and mobility analysis · Cross-national policy analysis · Governance case studies
AI & Applied Research
Machine learning and deep learning for spatial modelling · Geospatial AI · Climate and environmental risk assessment

PROFESSIONAL ASSOCIATIONS

- American Association of Geographers
- Economic Geography Research Group
- Quantitative Methods Research Group
- Transport Geography Research Group
- Urban Geography Research Group

HONORS, AWARDS & SERVICE

- Presenter, Korea International Model UN 2024, DISEC Committee, Seoul
- Global Peace Leadership Corps (Seoul, 2024)
- Government of Korea Scholarship recipient
- AKS International Young Scholars' Seminar
- Certification: TOPIK
- Executive Secretary, ISRK (2023-2024)

LANGUAGES

- English - Native
- Hindi - Professional Working Proficiency
- Korean - Professional Working Proficiency